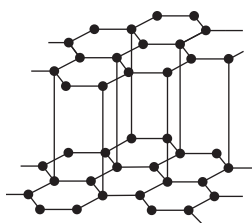
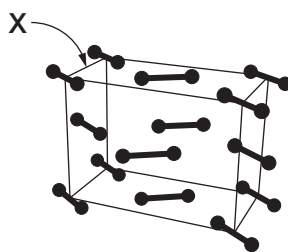
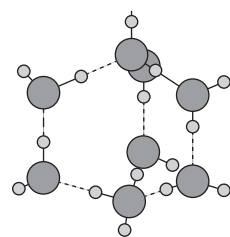
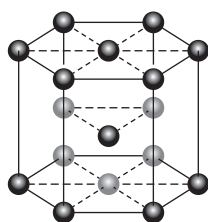
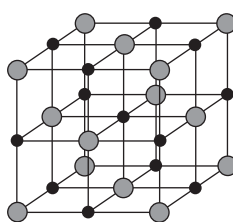
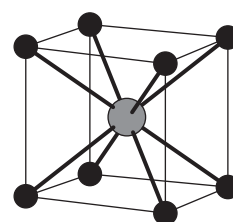


SECTION BAnswer **all** questions.

8. (a) The diagrams below represent the structures of cadmium metal, caesium chloride, graphite, ice, iodine and sodium chloride.

**A** graphite**B****C****D** cadmium**E****F**

- (i) Label the remaining structures in the spaces provided.

[2]

- (ii) Complete the table showing the trend in melting temperatures of the six substances.

[2]

Substance	Melting temperature
.....	Lowest
.....	
.....	
caesium chloride	
sodium chloride	
.....	
	Highest



- (iii) For structure **B**, name the type of bond or force represented by the letter **X**. [1]

.....

- (iv) Name the **two** substances which are good electrical conductors in the solid state. [1]

.....

.....

- (v) Explain why graphite is suitable for use in pencils. [2]

.....

.....

.....

- (vi) Cadmium is a typical metal.
Give a brief description of metallic bonding.
You may include a diagram to support your answer. [3]

.....

.....

.....

.....

.....



- (b) Many houses have been built on disused industrial sites. A housing developer wants to test whether the soil on a particular site is contaminated with cadmium ions (Cd^{2+}).

They extract a sample and prepare a solution. Cadmium ions behave in the same way as magnesium ions when treated with sodium hydroxide solution and sodium sulfate solution.

- (i) State what the developer would see if cadmium ions were present when they added:

I. sodium hydroxide solution. [1]

.....

II. sodium sulfate solution. [1]

.....

- (ii) Write an ionic equation, including state symbols, for **one** of the above observations. [1]

.....

